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Differentiation of Binary Cross Entropy (BCE)

Loss function:

Step for a single learning sample,

$$L = -[y \log(a) + (1-y) \log(1-a)]$$

where,

y = actual level (0 or 1)

a = predicted probability from sigmoid activation

Differentiating both side of the BCE function, with respect to a ,

$$\frac{dL}{da} = - \left[\frac{y}{a} + \frac{1-y}{1-a} \right]$$

$$= -\frac{y}{a} - \frac{1-y}{1-a}$$

$$= \frac{-y + ay + a - ay}{a(1-a)}$$

$$= \frac{a-y}{a(1-a)}$$

$$\text{Hence, } \frac{dL}{da} = \frac{a-y}{a(1-a)}$$

For sigmoid activation, $a = \sigma(z)$

$$\frac{da}{dz} = a(1-a)$$

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Using the chain rule,

$$\frac{dl}{dz} = \frac{dl}{da} \cdot \frac{da}{dz}$$

$$= \left(\frac{a-y}{a(1-a)} \right) \cdot a(1-a)$$

$$= a-y$$

Therefore, $\frac{dl}{dz} = a-y$

Final result, $S = \frac{dl}{dz} = a-y$

$$\left[\frac{b-1}{a-1} + \frac{b}{a} \right] - = \frac{b}{ab}$$

$$\frac{b-1}{a-1} + \frac{b}{a} =$$

$$\frac{b(a-1) + b(a-1)}{(a-1)a}$$

$$\frac{b(a-1)}{(a-1)a}$$

$$\frac{b(a-1)}{(a-1)a} = \frac{b}{ab}$$

$$\frac{b(a-1)}{(a-1)a} = \frac{b}{ab}$$